Book 1

What This Actually Is

An orientation to Live Actions Genius

Genius is an AI assistant wired directly into a running copy of Ableton Live. It can read your Set, write real MIDI into it, load and configure devices, drive the transport, and measure your audio — through 128 purpose-built actions rather than by pretending to type at your keyboard. This book explains what that means in practice, what the moving parts are, and how to get a useful answer out of it on the first try.

In this book

- How the pieces fit together, and what breaks when one is missing
- The three things Genius can do that a normal chatbot cannot
- Where its music knowledge comes from, and why there are two sources
- Choosing a chat model for the job in front of you
- The habits that separate good results from frustrating ones

Built against Live Actions v1.0.0. Action surface extracted directly from the shipping code. Behaviour verified against Ableton Live 12.4.15b3.

The one-paragraph version

You type a request in plain English. Genius decides which of its 128 actions to call, in what order, with what arguments. Those calls travel over a local bridge into a Max for Live device sitting in your Set, which talks to Ableton through Live's own scripting interface. Real things change in your project. Genius reads the results back and tells you what happened. Your audio and project files stay on your machine. What you type, and what the actions read back from your Set and the knowledge base, is sent to the AI provider whose API key you entered, as part of the conversation.

The moving parts

Five things have to be true for a Live action to work. When something is broken, it is almost always one of these, and they are listed in the order worth checking.

Part	What it does	How you know it's wrong
Ableton Live	Hosts your project and does the actual work. Live 12 with Max for Live.	Nothing responds at all; live_connect fails immediately.
The bridge device	A Max for Live device in your Set. Without it loaded, Live cannot hear the app, no matter how healthy everything else is.	live_connect fails but Live is clearly running and fine.
Live Actions	The desktop app. Holds your settings, ports, folder paths and API keys.	Settings look right but the app can't reach the bridge — check the port pair.
A chat model	The brain choosing which actions to call. Needs your own API key from xAI, OpenAI, Google or Anthropic, entered in Preferences. Your provider bills you for use.	Actions run but the choices are poor, or it narrates work it never did.
The knowledge base	A bundled database of 8 datasets, plus a Knowledge folder for your own notes and manuals. Genius searches these before answering from memory.	Theory answers are vague or it says it has no source for something it should know.

If only one thing sticks

The bridge device has to be present **in the Set you have open**. It is not a global setting. Open a different project and the device is not there until you add it — which is the single most common reason Genius appears to have stopped working.

What makes this different from a chatbot

Any AI can describe a chord progression. Three things here are genuinely different in kind, not just degree.

1. It changes your actual project

When Genius writes a bassline, real MIDI notes appear in a real clip that you can then edit by hand. It is not producing a description, a MIDI file to import, or a suggestion. The change has already happened in Live by the time it finishes replying.

2. The music comes from real algorithms, not improvisation

Euclidean rhythms use Bjorklund's algorithm. Key detection uses Krumhansl-Schmuckler. Chord voicings follow actual voice-leading rules. The language model decides *which* generator to call and with what settings — it does not invent notes one at a time. That is why the same request twice gives you the same pattern twice, and why the output is musically coherent rather than plausible-sounding.

3. It can measure what it did

Genius can capture Live's output and analyse it for real: LUFS, true peak, crest factor, spectrum. That makes a closed loop possible — change something, measure it, adjust based on the result. This is the one capability that feels like an assistant rather than a tool, and it is covered in Book 2.

The 128 actions at a glance

Book 5 documents every one of these in full. This is the shape of the toolset, so you know roughly what exists before you go looking.

Family	Count	What it covers
Connect & Read Live State	7	Start here.
Transport, Tempo & Global Feel	22	Everything that affects the whole Set rather than one track: playback, tempo, swing, groove, recording modes and the global mixer controls.
Tracks & Mixer	8	Creating, reading, renaming and reshaping tracks, plus the per-track mixer.
Scenes	4	Session View rows — reading, editing and capturing them.
Session Clips — Create & Edit	15	The clip-level workhorses: making clips, writing notes into them, and shaping how they loop, warp and launch.
Clip Automation & Envelopes	4	Drawing parameter movement inside clips, procedurally.
Arrangement Timeline	10	Everything on the horizontal timeline: placing clips, reading and writing their notes, moving them around, and recording into them.
Take Lanes & Comping	4	Multiple takes stacked under one track, for comping the best parts together.
Devices, Racks & Macros	13	Instruments and effects: inspecting them, setting their parameters, loading them, and wrangling Racks and Drum Racks.
Algorithmic & Generative MIDI	8	Deterministic music generation.
MIDI Editor Note Selection	2	Added in v1.
Genre Recipes	2	Pre-built multi-step production templates that run as a single call.
Audio Analysis, Recording & Mix	5	Genius's ears.
Samples & Browser	5	Finding sounds on disk and in Live's own library.
Vocals & Audio Files	2	Two actions that run on the desktop app itself rather than inside Live.
Knowledge Base	3	Where Genius's music knowledge comes from.
Standing Instructions	1	Your own house rules for Genius, saved as presets and attached to a Live Set.
Offline Project Files (.als)	5	Reads Ableton project files straight off disk.
Diagnostics, Views & Escape Hatches	8	The tools for when something isn't working, plus direct access to Live's own API for anything the purpose-built actions don't cover.
Total	**128**	

Where its knowledge comes from

There are two separate local sources, and Genius is instructed to try them in a strict order. Knowing which one an answer came from tells you how much to trust it.

Source	Contents	When it's used
Music/Audio database `library_db_search`	8 bundled datasets: beat-making recipes by genre, preset-rack breakdowns, mix engineering figures, Vital sound design, beginner music theory, notes on Ableton remote-control projects. You can add your own.	**Always first.** Fast indexed search over material that was deliberately curated and structured.

Source	Contents	When it's used

Knowledge folder
`library_search`
`library_read`
	Your own files: notes, manuals you own, exports. It ships empty; drop .md, .txt or .pdf files into it.	**Fallback**, when the database has nothing — typically a named third-party plugin or a specific piece of hardware you have a manual for.

Both live on your own drive and are searched locally. The passages a search finds are then sent to your AI provider so the model can read them. The router document that decides between sources, `INDEX.md`, ships with the app; put your own INDEX.md in the Knowledge folder to replace it.

Choosing a model

Genius runs on whichever chat model you configure in Settings, and the choice matters more than people expect. Driving six dependent tool calls in the right order is a genuinely different skill from explaining modal harmony well. Every prompt in Book 3 carries a suggested tag.

Tag	Best at	Reach for it when
CLAUDE	precise multi-step Live automation / exact tool sequencing	The job is multi-step, touches several tracks, or must not go wrong.
GPT	balanced generalist, strong second choice for automation	You want a solid all-rounder, or Claude is unavailable.
GROK	long-form theory and creative coaching, big-context digestion	You want teaching, theory, or a long discursive answer.
GEMINI	fast, cheap single-fact lookups and simple citations	You want one fact, fast and cheap.
ANY	simple enough that model choice barely matters	It's a single obvious action with no ambiguity.

Getting good results

Let it look before it leaps

The highest-leverage habit there is. “Read my Set, then…” costs one action and prevents most of the failures in this guide. A model that has read your project makes dramatically better choices than one guessing at its shape.

Name things, don't number them

Track numbers are positions as Live displays them — a track *called* “Track 3” may sit in position 7, and digits in a name mean nothing. Say “the track called Bass”, or ask what's selected and say “this one”.

Say where output should go

“Put the variation in slot 4” protects the original. Without it you may get the source overwritten, which is undoable but annoying.

Ask it to check unfamiliar parameters

“Tell me the range first” makes it call `live_read_param_range` instead of assuming a filter cutoff runs 0–1 when it actually runs 0–127.

Verify anything that matters

A model can produce a confident summary of work it did not do. Snapshot before, diff after, or simply re-read the thing it claims to have changed. Believe the read, not the summary. This is the most important habit in the entire guide.

Break big builds into checkpoints

A twelve-step build that goes wrong at step four spends the next eight steps compounding the error. Stop and look at stages you can actually verify.

Where to go next

Book 2 makes the case for what to actually try. **Book 3** is the prompt library — paste-ready, tagged by difficulty and model. **Book 4** is the honest list of what this cannot do, and is worth reading *before* you hit a limit rather than after. **Book 5** documents all 128 actions individually.